

Q4) Write a C program for polyalphabetic substitution cipher using a separate monoalphabetic substitution cipher for each successive letter of plaintext, depending on the key.

Aim: To write a C program to implement the **Polyalphabetic Substitution Cipher (Vigenère Cipher)**, where each letter of the plaintext is encrypted using a corresponding letter of a repeating keyword.

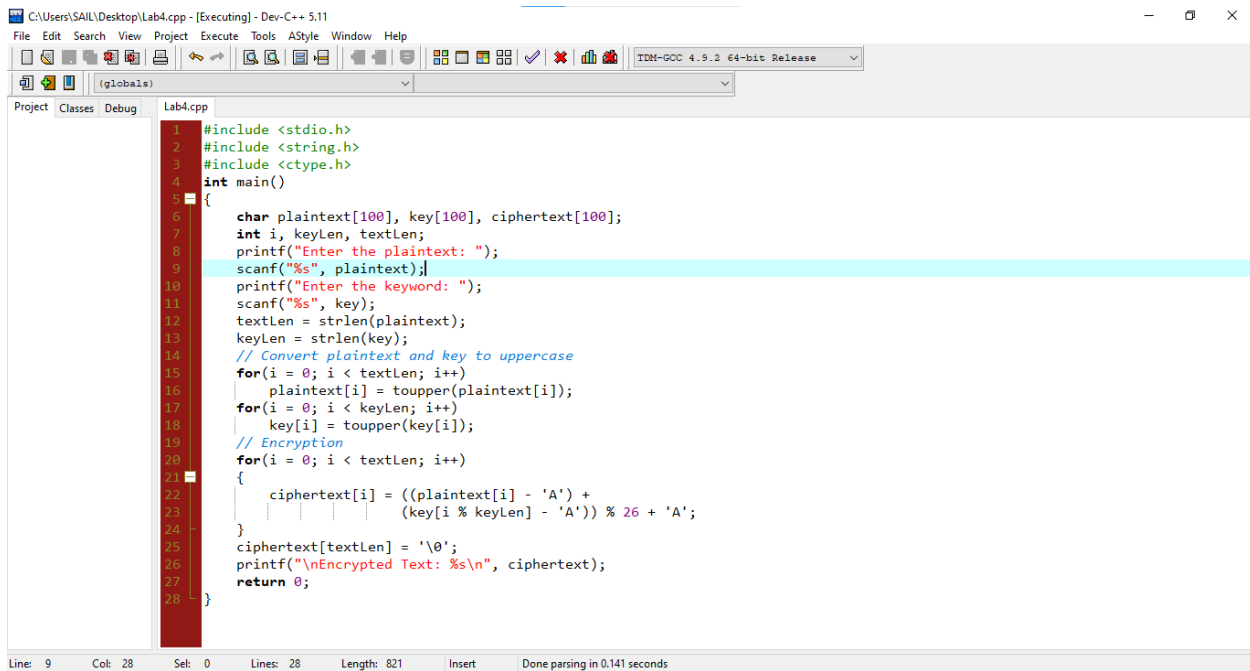
Algorithm:

1. Start the program.
2. Declare character arrays for the plaintext, keyword, and ciphertext.
3. Read the plaintext and keyword from the user.
4. Convert all characters of the plaintext and keyword to uppercase.
5. Repeat the keyword until it matches the length of the plaintext.
6. For each plaintext character, convert both the plaintext and keyword characters into numerical values ($A = 0, B = 1, \dots, Z = 25$).
7. Encrypt each character using the formula: **Cipher = (Plaintext + Keyword) mod 26.**
8. Convert the encrypted numerical value back to its corresponding alphabet and store it in the ciphertext.
9. Display the encrypted ciphertext.
10. Stop the program.

Result:

Thus, the C program to implement the **Polyalphabetic Substitution Cipher (Vigenère Cipher)** was successfully executed, and the corresponding encrypted ciphertext was obtained.

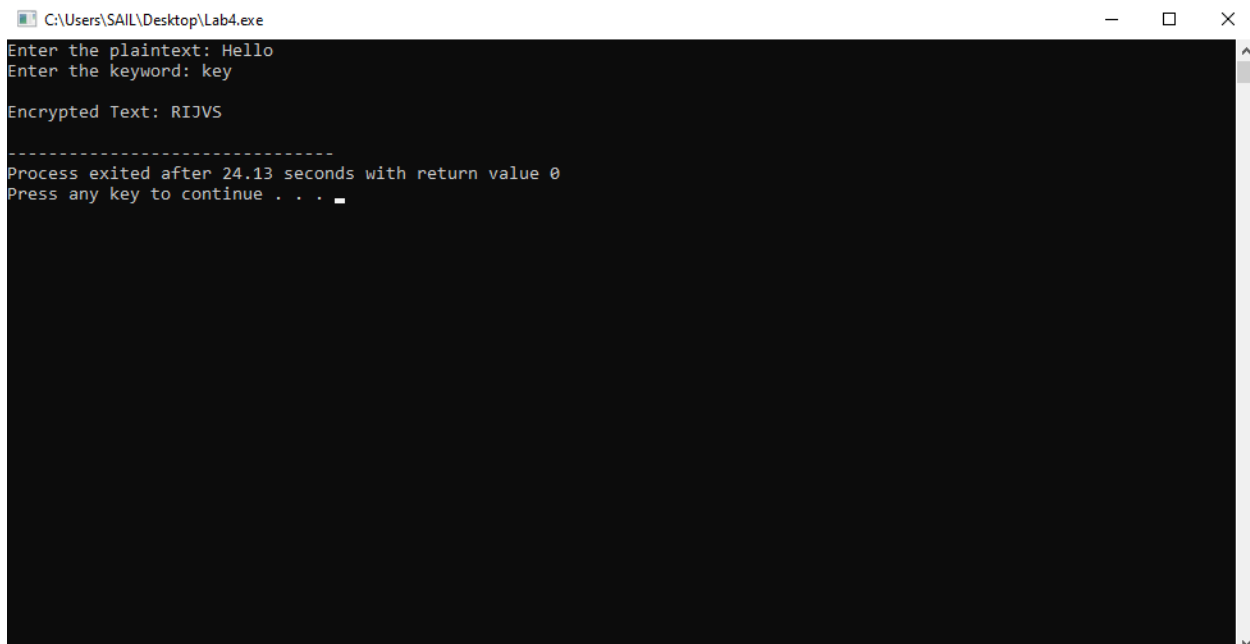
Code:



```
1 #include <stdio.h>
2 #include <string.h>
3 #include <ctype.h>
4 int main()
5 {
6     char plaintext[100], key[100], ciphertext[100];
7     int i, keyLen, textLen;
8     printf("Enter the plaintext: ");
9     scanf("%s", plaintext);
10    printf("Enter the keyword: ");
11    scanf("%s", key);
12    textLen = strlen(plaintext);
13    keyLen = strlen(key);
14    // Convert plaintext and key to uppercase
15    for(i = 0; i < textLen; i++)
16        plaintext[i] = toupper(plaintext[i]);
17    for(i = 0; i < keyLen; i++)
18        key[i] = toupper(key[i]);
19    // Encryption
20    for(i = 0; i < textLen; i++)
21    {
22        ciphertext[i] = ((plaintext[i] - 'A' +
23                        (key[i % keyLen] - 'A')) % 26 + 'A');
24    }
25    ciphertext[textLen] = '\0';
26    printf("\nEncrypted Text: %s\n", ciphertext);
27    return 0;
28 }
```

Line: 9 Col: 28 Sel: 0 Lines: 28 Length: 821 Insert Done parsing in 0.141 seconds

Output:



```
C:\Users\SAIL\Desktop\Lab4.exe
Enter the plaintext: Hello
Enter the keyword: key

Encrypted Text: RIJVS

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Process exited after 24.13 seconds with return value 0
Press any key to continue . . .
```